

Using Ecosampler – an EwE uncertainty sampler

Ecosampler is a light-weight utility for the Ecopath with Ecosim that records and runs alternate balanced Ecopath parameter sets, referred to as 'samples', to test parameter uncertainty.

Installation

Although its core logic is included with the EwE libraries, the Ecosampler plug-in is needed to interact with the graphical user interface. The plug-in can be obtained on request to ewedevteam@gmail.com.

Interface

The EcoSample interface is launched by choosing "Navigator > Tools > Ecosampler". The Ecosampler interface shown in Figure 1 offers the following elements:

1: A tool bar with the following facilities:

A record button to toggle sample recording (on/off).

Import facilities to import samples stored elsewhere.

2: A list of recorded samples, with for each sample, a sequential number, the data and computer system the sample was recorded on, and whether a sample is currently loaded in the EwE user interface.

3: Controls to load or delete selected samples.

4: Controls to run a select number of samples through EwE.

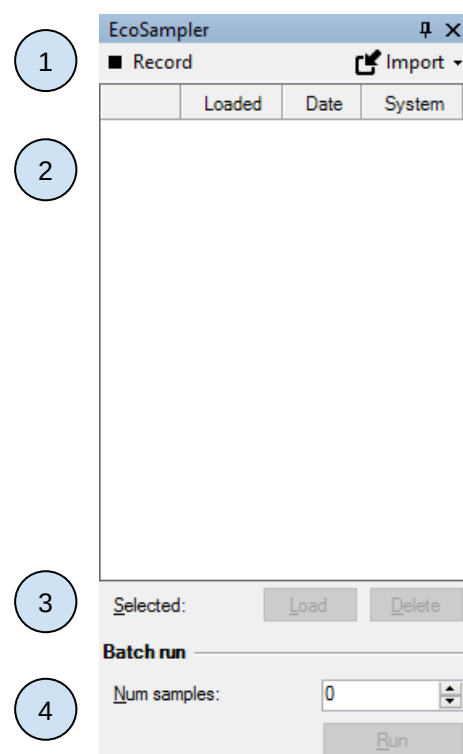


Figure 1 - Ecosampler interface:
1) toolbar, 2) Samples list, 3) Sample management, 4) Run panel

Recording samples

To record samples make sure that the "Record" button is pressed (see Figure 2). The record button is red when active, and black when inactive.

While recording, Ecosampler listens to the execution of the EwE Monte Carlo routine, and records any alternate mass-balanced model found by Monte Carla as a "sample" in the currently loaded model database.

Managing samples

Ecosampler offers the possibility to load a stored sample in the EwE user interface to assess its validity. This will load the alternate parameter set as computed by Monte Carlo into the Ecopath input parameters, where they can be used to run any EwE module.

The Ecosampler user interface indicates which sample is loaded, and offers the ability to restore the original EwE parameter set. This is reflected in Figure 2, where sample 9 is shown as loaded, and the option to unload the sample is now offered.

Users can delete samples that provide unrealistic alternate models.

⚠ Note that Ecosampler will prevent users from accidentally overwriting their model inputs with a loaded sample. If a sample is loaded when users try to save the model, Ecosampler will first restore the original parameter set before the save operation is executed. To save the loaded sample configuration please use “Menu > File > Save as...” to store the model under a new name.

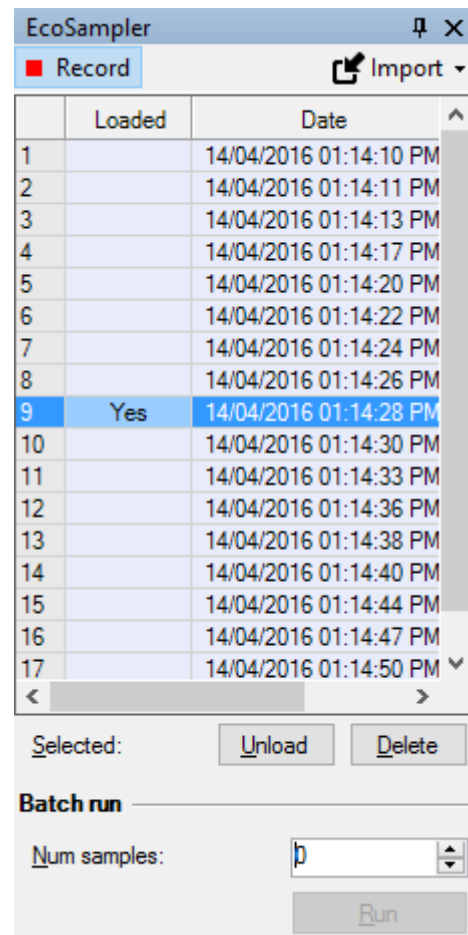


Figure 2 - Recording samples

Importing samples

Especially when perturbing the diet composition matrix, the Monte Carlo routine may require a long time to find an alternate model. Running multiple instance of EwE to record samples on different computers - or on the same computer - will greatly speed up the recording process. You can then use the 'Import samples' functionality to import samples from another model database to bring all samples together.

⚠ Note that Ecosampler will import samples from any model. The responsibility to check whether models match is the responsibility of the user.

⚠ Note that Ecosampler is also intended to import samples generated and stored by the Cefas MSE routines. This logic has not been implemented yet.

Sample validity

Samples are alternate mass-balanced configurations for a given Ecopath model. If the Ecopath parameter input parameters are changed and these changes are saved, any stored sample is in principle invalid. The source code contains provisions to ask the user if samples should be deleted, but perhaps this should be left to the user without Ecosampler trying to be too smart.

Running samples

When the Run button is clicked, Ecosampler first checks if Ecosim, Ecospace and Ecotracer were loaded at the time the batch run is stated. Ecosampler will execute the current model configuration as a baseline sample, and will store this in a subfolder “Sample_baseline” under the default core output path. If Ecopath does not balance the user is notified and the batch run is aborted.

If Ecopath balanced, then Ecosampler will start processing the list of samples, starting at the first stored sample (with index 1) up to the end of the list or until the number of samples listed in the “Number of samples to run” has been processed. It will run each sample in turn.

Running a sample entails the following steps:

- Change the core output path by adding “Sample_XXXXX” to make sure that auto-saving components store their data organized per sample.
- Load the sampled parameter set into Ecopath and run Ecopath.
- If Ecopath balances, any Ecosim and Ecospace configuration present at the start of the batch run is also executed for the sample.

 Note that Ecosampler is a simple tool that only reruns the present EwE configuration. Ecosampler has no means to decide which data may need saving, and has no means to control which plug-ins will run. Instead, it relies on the current configuration of the EwE software to take care of this. For instance, if the Ecological Indicators plug-in is set to run with Ecopath and Ecospace, and Ecospace is set to auto-save ASCII output maps in Ecospace parameters, these settings will also apply to every sample ran by Ecosampler.

Using the Ecosampler output

The output of Ecosampler is a unique folder for each sample under the EwE output directory. Each sample folder contains the various output files written by EwE components set to Autosave their results when ran with each sample.

Troubleshooting

Ecosampler activities are written to the Ecopath log file, which can be viewed by choosing “Main Menu > About > View Log File”.

Further development

A series of future changes can be foreseen:

- Prompt user when deleting a sample because delete cannot be undone
- Hide “System” column when there are no samples from another computer
- Implement ratings
- Automatically rate samples when generated?
- Implement Cefas MSE import